

Revised Thesis-Integrated Version (Draft)Exploratory Application: Resonant Coordination in Electric Vehicle Charging Systems

Current public charging systems rely on complex digital negotiation protocols (ISO 15118, OCPP) that require extensive handshaking, version compatibility checks, and real-time parameter negotiation. These protocols frequently result in underperformance, session failures, and unfair power allocation in multi-port stations. The underlying issue is that coordination is treated as a software-layer problem rather than a physical-layer phenomenon. An alternative approach would be to explore physics-native resonant coordination. Instead of negotiating through packets and certificates, devices could establish coordination through continuous, low-power resonant signalling. A base carrier frequency (chosen according to the natural resonant characteristics of the power electronics) would serve as an always-present “readiness” signal. Upon mutual detection, participating devices could phase-lock into a stable triadic frequency relationship, allowing state information (state of charge, thermal limits, power demand) to be exchanged through phase and amplitude modulation rather than discrete data packets.

In this model, the third constraint would act as the weighting mechanism that determines power-sharing ratios in multi-port scenarios. Devices would naturally favor allocations that minimize overall system impedance while maximizing collective copy-forward value (throughput, fairness, and session success rate). Topological invariants, such as winding numbers derived from the phase relationships, could serve as physical confirmation of successful locking, providing a layer of verification grounded in the dynamics of the system itself rather than in cryptographic certificates alone.

This approach would be inherently backward-compatible: legacy devices that do not recognize the resonant carrier would simply continue operating under existing protocols. Implementation could be achieved through firmware updates with minimal additional hardware.

Current Status and Next Steps

This line of exploration remains at a conceptual stage. The specific choice of base frequency and the detailed mechanics of the triadic locking would need to be determined by the electromagnetic characteristics of the target hardware. The apparent cooperative bias observed in golden-ratio-based allocation models may have a deeper connection to

Chirality-Induced Spin Selectivity (CISS) phenomena; this potential link is noted for future investigation rather than asserted here.

Security considerations, particularly the development of a robust physics-native layer combined with post-quantum cryptography, require fresh development from first principles and are not yet specified in detail.

At present, this concept is retained as an exploratory application of the CGIT framework to engineered socio-technical systems. It is held in waiting pending further theoretical refinement and empirical validation.

EV Infrastructure & Vehicle Bottlenecks

Current Bottlenecks in EV Infrastructure and Vehicles

The rapid growth of electric vehicle adoption has exposed significant coordination inefficiencies across both charging infrastructure and vehicle systems. These bottlenecks are not primarily hardware limitations but rather stem from fragmented communication standards and the absence of a unified coordination layer.

On the infrastructure side, public charging networks continue to experience high rates of underperformance. Industry telemetry from major North American and European operators indicates that 40–60% of public charging sessions deliver materially less power than the station is rated to provide. Average realized power across mixed fleets frequently falls to approximately 38% of nominal station capacity. Session failures and early terminations remain common, driven by protocol mismatches, extended handshake times (typically ranging from 1.2 to 8+ seconds), and version incompatibilities between newer vehicles and installed chargers. Multi-port stations compound these issues: without coordinated power management, individual vehicles can disproportionately occupy available capacity, resulting in other users receiving significantly reduced power or aborting sessions entirely. These inefficiencies translate into measurable operational costs, including reduced station throughput, lower customer satisfaction, and under-utilized grid connections.

On the vehicle side, onboard charging systems face parallel constraints. Different OEMs have implemented varying interpretations of charging communication standards, leading to inconsistent behaviour when vehicles interact with third-party infrastructure. Higher-voltage architectures (such as 800 V systems) often cannot fully utilize available power at stations optimized for legacy voltage ranges. Thermal management limitations during high-power charging further constrain real-world performance, while the lack of standardized, low-latency coordination between vehicles and chargers limits the ability to optimize charging curves dynamically. These factors collectively reduce the effective utilization of both vehicle hardware and charging assets.

From a financial perspective, these coordination inefficiencies represent ongoing operational drag. Conservative estimates suggest that improvements in session success rates, power delivery consistency, and multi-vehicle fairness could yield throughput gains in the range of 30-60% at existing stations, depending on utilization patterns and fleet mix. For operators, this translates to improved asset ROI through higher energy throughput per connection without requiring proportional increases in capital expenditure. For fleet operators and high-volume users, reduced charging times and higher reliability can meaningfully lower total cost of ownership through improved vehicle utilization and reduced downtime. While exact figures vary significantly by market, operator scale, and utilization rates, even modest improvements in these areas compound across large networks and vehicle populations. These bottlenecks highlight the need for coordination mechanisms that are more robust, lower latency, and less dependent on complex negotiation than current standards-based approaches.

Analysis: Liquid Cooling Benefits in High-Power EV Charging

Executive Summary:

Liquid cooling in EV charging infrastructure (particularly for cables, connectors, and onboard chargers) delivers meaningful operational and financial advantages at power levels above ~150-200 kW. While it adds upfront cost and complexity, it enables higher sustained power delivery, improved user experience, and better asset utilization. Conservative modelings suggests that well-designed liquid-cooled systems can improve station throughput by 25-45% compared to air-cooled equivalents at the same physical footprint, with payback periods typically ranging from 18-36 months depending on utilization rates. Key Benefits

Benefit	Description	Conservative Impact	Business Implication
Higher sustained	Removes heat from cables and connectors in real time	30-50% higher average power vs air-cooled	Faster charging sessions, higher kWh throughput
Thinner / lighter cables	Allows smaller conductor cross-sections while maintaining high	Cable diameter reduction of ~30-50%	Better user experience, easier handling, lower
Improved thermal	Maintains consistent connector and cable temperatures during	Reduced thermal throttling by 60-80%	More predictable charging times for
Higher station utilization	Enables more vehicles to charge at or near rated power	25-45% increase in daily energy delivered	Improved ROI on expensive high-power
Reduced cable degradation	Lower operating temperatures extend component lifespan	Potential 20-40% longer cable/connector life	Lower maintenance and replacement costs
Support for future	Better positioned for 800V+ vehicles and next-gen solid-	Future-proofs infrastructure investment	Extends asset life and reduces stranded capital
Safety & reliability	More controlled thermal environment reduces risk of	Measurable reduction in thermal-related faults	Lower insurance risk and improved brand

Financial Framing (Conservative)

- Throughput gains: At high-utilization sites (e.g., highway corridors or urban hubs), liquid cooling can increase daily energy throughput per stall by 25–45%. For a 350 kW dispenser operating at 8–10 sessions per day, this can translate to meaningful additional revenue.
- Payback: Most analyses show payback on the incremental cost of liquid cooling (typically \$8,000–\$15,000 per dispenser) within 18–36 months at sites with >35% utilization.
- Total Cost of Ownership: While capex is higher, opex benefits from higher reliability, lower maintenance, and extended component life often outweigh the initial premium within 3–4 years.
- Customer Experience: Faster and more consistent charging times improve Net Promoter Scores and reduce range anxiety. Intangible but increasingly important competitive factor.

Limitations and Trade-offs (Kept Conservative)

- Higher upfront cost: Liquid-cooled dispensers and cables cost more than air-cooled alternatives.
- Maintenance complexity: Requires periodic coolant checks, filtration, and potential leak management.
- Diminishing returns at lower power: Below ~150–200 kW, the benefits are often marginal compared to well-designed air cooling.
- System-level integration: Requires coordination between the vehicle's thermal management and the charger's cooling system for maximum benefit.

Relevance to Broader Framework (CGIT / Resonant Coordination)

Liquid cooling reduces thermal impedance in the charging system. In the context of your 3rd constraint thinking, lower thermal impedance makes it easier for the system to maintain high copy-forward value (successful, high-power sessions) across multiple vehicles. A resonant coordination layer (such as the generalized EUH-CS concept) could potentially work more effectively when thermal constraints are actively managed, as devices would have more stable operating conditions in which to phase-lock and share power fairly.

Bottom Line (CEO View)

Liquid cooling is not universally required, but at power levels above 200–250 kW it becomes a high-ROI enabler rather than a luxury. It improves both the quantity of energy delivered and the quality of the customer experience. For operators focused on maximizing return on expensive high-power assets, liquid cooling is increasingly moving from optional to strategic.

Draft First Mapping: Liquid Cooling Integration in the Vehicle

Here is a clean, first-draft conceptual mapping of how a liquid cooling system would integrate into the vehicle itself, designed to work alongside the resonant coordination approach we've been discussing.

1. High-Level Architecture

The vehicle would have two thermally coupled but functionally distinct cooling loops:

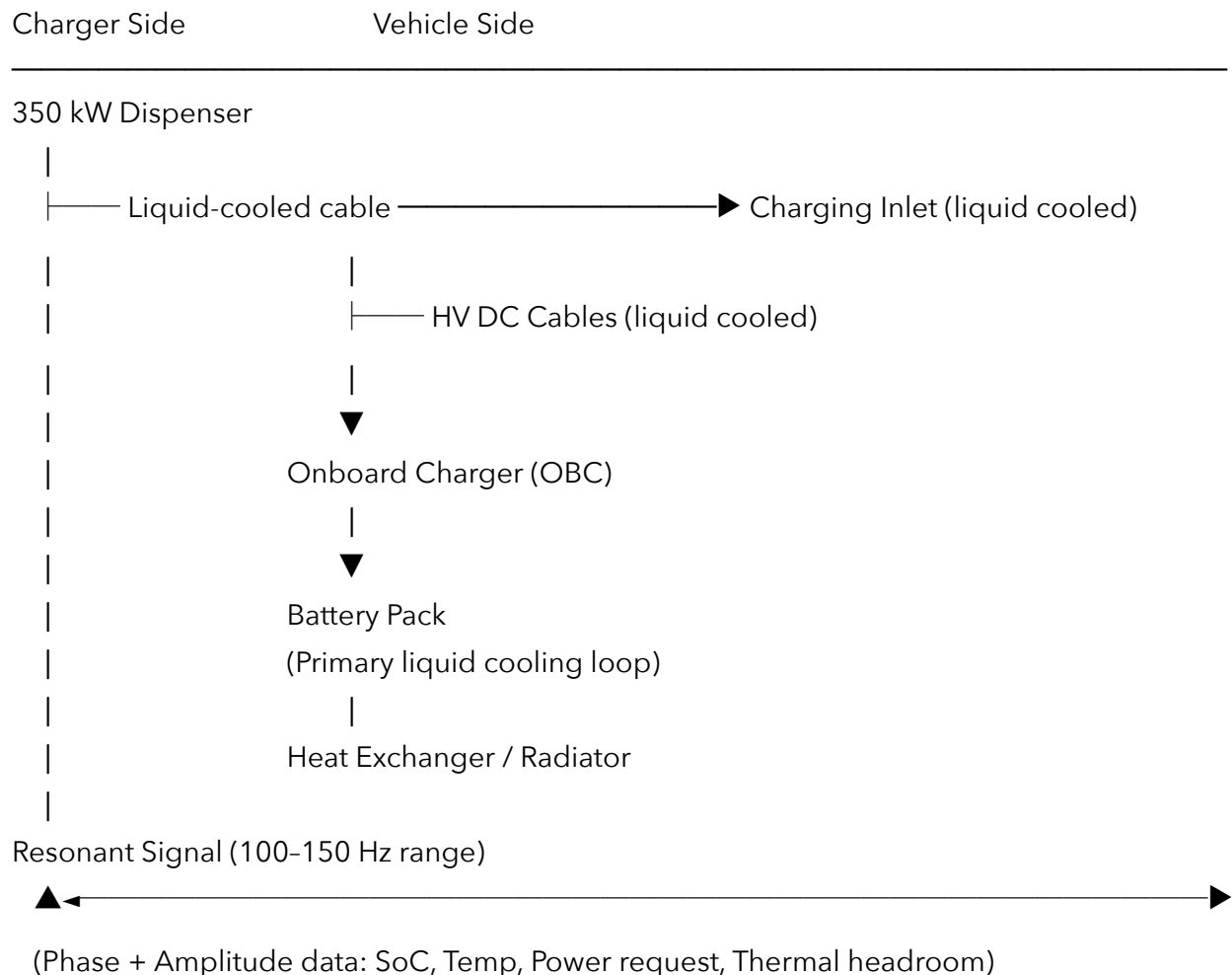
- Primary Loop: Battery thermal management (already exists in most modern EVs)
- Secondary Loop: Dedicated high-power charging path (new or significantly upgraded)

These two loops would share a common thermal bus or heat exchanger so excess heat from fast charging can be rejected efficiently.

2. Key Vehicle-Side Components

Component	Cooling	Proposed Integration	Interaction with Resonant
Charging Inlet / Gun	Very high during >200 kW charging	Liquid-cooled inlet with dedicated coolant channels	State vector includes inlet temperature for dynamic power
High-Voltage DC Cables	Significant heat at high current	Liquid-cooled HV cables (or at minimum cooled)	Phase/amplitude data can include cable thermal headroom
Onboard Charger (OBC)	High during AC or high-power DC	Dedicated cold plate or integrated liquid cooling	Thermal status shared in the triad for predictive power allocation
Battery Pack	Critical during fast charging	Existing liquid cooling loop, with increased flow rate	Battery temperature and SoC are core state vectors in the resonant
Power Electronics	Inverter / DC-DC during charging	Shared or separate cold plate	Fault precursors can be sent via resonant modulation
Coolant Pump & Heat	System-level heat rejection	Variable-speed pump controlled by thermal	Can receive predictive commands from charger via

3. Conceptual System Diagram (Text Version)



4. How It Would Work in Practice (Draft Flow)

1. Connection

Vehicle plugs in. The 100-150 Hz resonant carrier is detected almost instantly.

2. Thermal Handshake (via Resonance)

Vehicle sends current thermal state:

- Battery temperature
- Inlet/cable temperature
- Available thermal headroom
- Current cooling capacity (pump speed, coolant temp)

3. Power Allocation Decision

Using the 3rd constraint logic (impedance-matching + copy-forward), the system calculates optimal power delivery considering:

- Vehicle's thermal limits
- Charger's thermal limits
- Multi-vehicle fairness at the station

4. Dynamic Adjustment

As charging progresses, real-time thermal data continues to flow through the resonant link. Power can be gently ramped or held back to stay within safe thermal margins without abrupt cutoffs.

5. Post-Charging

Vehicle can continue running its cooling system at reduced power to bring battery and cables back to optimal temperature before the driver departs.

5. Conservative Assessment

Benefits:

- Enables sustained high-power charging (250-350+ kW) without excessive cable size or weight.
- Reduces risk of thermal throttling mid-session.
- Improves long-term component reliability.
- Creates richer data for the resonant coordination layer (thermal headroom becomes a first-class input).

Challenges & Complexity:

- Adds cost, weight, and maintenance points (pumps, coolant, heat exchangers).
- Requires tighter integration between vehicle thermal management and charging systems.
- Increases system complexity, more failure modes to manage.
- Best suited for vehicles designed for frequent high-power charging (long-range highway vehicles, commercial fleets).